

・微分の感覚を掴む

$\{f(x)^a\}' = a \cdot f(x)^{a-1} \cdot f'(x)$	$\{\sqrt{f(x)}\}' = \frac{f'(x)}{2\sqrt{f(x)}}$
$\{\sin f(x)\}' = \cos f(x) \cdot f'(x)$	$\{\cos f(x)\}' = -\sin f(x) \cdot f'(x)$
$\{\tan f(x)\}' = \frac{f'(x)}{\cos^2 f(x)}$	$\{\frac{1}{\tan f(x)}\}' = -\frac{f'(x)}{\sin^2 f(x)}$
$\{\log f(x)\}' = \frac{f'(x)}{f(x)}$	$\{\log_a f(x)\}' = \frac{f'(x)}{f(x) \cdot \log a}$
$\{\log f(x) \}' = \frac{f'(x)}{f(x)}$	$\{\log_a f(x) \}' = \frac{f'(x)}{f(x) \cdot \log a}$
$\{e^{f(x)}\}' = e^{f(x)} \cdot f'(x)$	$\{a^{f(x)}\}' = a^{f(x)} \cdot \log a \cdot f'(x)$

(例) 次の関数を微分せよ。

$$\begin{aligned}
 (1) \quad y &= (2x+1)(x^2-1)^2 & y' &= (2x+1)'(x^2-1)^2 + (2x+1)\{(x^2-1)^2\}' \\
 & & &= 2(x^2-1)^2 + (2x+1) \cdot 2(x^2-1)(x^2-1)' \\
 & & &= 2(x^2-1)^2 + 4x(2x+1)(x^2-1) \\
 & & &= 2(x^2-1)\{(x^2-1) + 2x \cdot (2x+1)\} \\
 & & &= 2(x^2-1)(5x^2+2x-1)_{,,}
 \end{aligned}$$

$$\begin{aligned}
 (2) \quad y &= \left(\frac{x-1}{2x+1}\right)^3 & y' &= 3\left(\frac{x-1}{2x+1}\right)^2 \left(\frac{x-1}{2x+1}\right)' \\
 & & &= 3\left(\frac{x-1}{2x+1}\right)^2 \cdot \frac{(x-1)'(2x+1) - (x-1)(2x+1)'}{(2x+1)^2} \\
 & & &= 3\left(\frac{x-1}{2x+1}\right)^2 \frac{1 \cdot (2x+1) - (x-1) \cdot 2}{(2x+1)^2} \\
 & & &= \frac{9(x-1)^2}{(2x+1)^4}
 \end{aligned}$$

$$\begin{aligned}
 (3) \quad y &= (x+1)\sqrt{x^2+1} & y' &= (x+1)'\sqrt{x^2+1} + (x+1)(\sqrt{x^2+1})' \\
 & & &= 1 \cdot \sqrt{x^2+1} + (x+1) \frac{(x^2+1)'}{2\sqrt{x^2+1}} \\
 & & &= \sqrt{x^2+1} + (x+1) \frac{2x}{2\sqrt{x^2+1}} \\
 & & &= \frac{2x^2+x+1}{\sqrt{x^2+1}}
 \end{aligned}$$

$$\begin{aligned}
 (4) \quad y &= \tan^3 2x & y' &= 3 \cdot \tan^2 2x \cdot (\tan 2x)' \\
 & & &= 3 \tan^2 2x \cdot \frac{(2x)'}{\cos^2 2x} \\
 & & &= \frac{6 \tan^2 2x}{\cos^2 2x} \quad ,,
 \end{aligned}$$

$$\begin{aligned}
 (5) \quad y &= \sin^2 x \cos 2x & y' &= (\sin^2 x)' \cos 2x + \sin^2 x (\cos 2x)' \\
 & & &= 2 \sin x (\sin x)' \cos 2x + \sin^2 x \cdot (-\sin 2x) \cdot (2x)' \\
 & & &= 2 \sin x \cos x \cos 2x - 2 \sin^2 x \sin 2x \\
 & & &= \sin 2x \cos 2x - 2 \sin^2 x \sin 2x \\
 & & &= \sin 2x (\cos 2x - 2 \sin^2 x) \quad ,,
 \end{aligned}$$

$$\begin{aligned}
 (6) \quad y &= \log(x + \sqrt{x^2+1}) & y' &= \frac{(x + \sqrt{x^2+1})'}{x + \sqrt{x^2+1}} \\
 & & &= \frac{1}{x + \sqrt{x^2+1}} \left\{ 1 + \frac{(x^2+1)'}{2\sqrt{x^2+1}} \right\} \\
 & & &= \frac{1}{x + \sqrt{x^2+1}} \left(1 + \frac{2x}{2\sqrt{x^2+1}} \right) \\
 & & &= \frac{1}{x + \sqrt{x^2+1}} \frac{\sqrt{x^2+1} + x}{\sqrt{x^2+1}} \\
 & & &= \frac{1}{\sqrt{x^2+1}} \quad ,,
 \end{aligned}$$

$$\begin{aligned}
 (7) \quad y &= e^{2x} \sin 3x & y' &= (e^{2x})' \sin 3x + e^{2x} (\sin 3x)' \\
 & & &= e^{2x} \cdot (2x)' \sin 3x + e^{2x} \cos 3x \cdot (3x)' \\
 & & &= e^{2x} \cdot 2 \cdot \sin 3x + e^{2x} \cos 3x \cdot 3 \\
 & & &= e^{2x} (2 \sin 3x + 3 \cos 3x) \quad ,,
 \end{aligned}$$